Nikolay Kurtov

Senior Software Engineer | Distributed Systems | Blockchain Protocols | Hardware Acceleration

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Summary

Seasoned Software Engineer with 15+ years of experience specializing in high-performance computing, large-scale distributed systems, and hardware acceleration (FPGA). Proven track record of leading technical initiatives at Google and NEAR Protocol, currently building Rialo, a real-world-ready L1, at SubZero Labs: verifiable off-chain execution, threshold cryptography, and zero-downtime protocol upgrades.

Experience

Senior Software Engineer — SubZero Labs, Rialo (July 2025 – Present)

- **REX (Real-world Execution):** Built the validator-side pipeline that lets on-chain programs call off-chain HTTP/WebSocket services deterministically: duty scheduling, replay from the commit index, result cache, RPC/CLI, metrics and tracing. Shipped a live stock-price demo on top of it.
- **Confidential REX:** Delivered threshold-encrypted requests end to end — **DKG** ceremony lifecycle (retries, disaster recovery, reshare), TEE-committee dispatch, partial-decrypt/combine — with a **Stateright** model of the key lifecycle; cut the encrypted pipeline latency pad **30s → 10s**.
- **Zero-downtime upgrades:** Own the no-re-genesis workstream: on-chain feature management (scheduled activation, two-step authority transfer), versioned persisted formats with golden-byte compatibility tests, epoch-keyed component config, and a nightly **rolling-upgrade e2e** from each deployed network to `main`.
- **Consensus & execution correctness:** Fixed epoch-handover liveness bugs (dying epochs restored on restart, Ready withheld for unrunnable epochs) and **WriteSet non-determinism** across restarts; split a 5k-line `bank.rs`; removed the inherited Solana `FeatureSet` machinery.
- **Specs & throughput:** Author of the **NORTHSTAR** contracts for REX, DKG, protocol upgrades, genesis and staking; ~800 merged PRs in 16 months.

System SW / Cryptography Engineer — Irreducible.com (2024 – June 2025)

- **Xlang Compiler:** Architected and implemented **Xlang**, a custom IR compiler for proprietary hardware cores. Achieved **99.5% hardware utilization** through advanced software pipelining and instruction scheduling.
- **FPGA Acceleration:** Delivered a **12x performance boost** for **Binius Sumchecks** and Zerochecks by offloading critical bottlenecks to FPGA (open-source link).
- **Team Leadership:** Mentored junior engineers to exploit fine-grained parallelism in Evalcheck, resulting in a **15x performance improvement**.
- **Production Deployment:** Designed the end-to-end software stack bridging high-level cryptographic protocols (Binius) with low-level FPGA drivers.

Tech Lead, Node Team — NEAR Protocol (2021 – 2023)

- **State Sync Re-architecture:** Led the complete redesign of the **State Sync mechanism**, significantly improving protocol reliability and node synchronization speed (design document).
- **Operational Excellence:** Reduced release-blocking incidents and improved the developer/validator experience by enhancing node monitoring with advanced metrics, structured logging, and **distributed tracing** (Grafana Tempo).
- **Core Contributions:** Maintained and optimized core blockchain protocol logic in **Rust**.

Senior Software Engineer / Tech Lead — Google (2011 – 2021)

- **Tech Lead, YouTube Core SRE (2.5 years):** Directed end-to-end reliability for YouTube's global infrastructure. Led the development of a C++ microservice dependency framework and established production readiness standards for core services.
- **Knowledge Graph Infrastructure & Quality (4 years):**
  - Optimized backend components for massive-scale graph data, significantly increasing throughput.
  - Improved ML models for music entity reconciliation, impacting **0.3% of global Google Search traffic**.
- **Data Quality & Personalized Offers (3.5 years):** Developed ML-driven systems for deduplicating business listings and built high-scale personalized offer distribution systems (Email/Web).

Early Career (2006 – 2011)

- **Intel (2008 – 2011):** Software Engineer / Intern. Optimized C++ compilers for Atom processors and developed distributed memory runtimes for Intel Concurrent Collections.
- **Intel Student Lab / xored / SWSoft (2006 – 2008):** Contributions to Apache Harmony compiler (edge profiling), Debian systems administration, and 3D game engine development.

Skills

- **Languages:** Rust (5+ yrs), C++ (14+ yrs), Python, Go
- **Expertise:** Distributed Systems, Blockchain Protocols, Threshold Cryptography & TEEs, High-Performance Computing (HPC), FPGA Acceleration, SRE & Scalability
- **Core Competencies:** System Architecture, Parallel Programming, Low-level Optimization, Cryptographic Engineering

Education

- **MS in Computer Science**, Novosibirsk State University (2009 – 2011). GPA: 5.0/5.0
- **BS in Computer Science**, Novosibirsk State University (2005 – 2009). GPA: 5.0/5.0

Programming Competitions

- **ACM ICPC World Finals:** 14th place (Harbin, 2010), 27th place (Orlando, 2011)
- **Yandex Open:** 11th place (Onsite round, 2010)

Hobbies

- **HEMA (Swordfighting):** Assistant Coach. Ranking
- **Running:** Marathon Personal Best: 3h 25m
- **Astronomy:** Built a DIY Solar Projection Telescope